

VEDANT BADVE

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PROFESSIONAL SUMMARY

Computer Engineering undergraduate (graduating 2027) with hands-on experience building machine-learning systems for healthcare diagnostics, security-focused anomaly-detection platforms, and full-stack engineering projects. Comfortable working across Python, cloud infrastructure (AWS), and core CS fundamentals, with a track record of leading cross-functional teams to top finishes in competitive hackathons. Seeking opportunities in Software Engineering, Machine Learning / AI, or Cybersecurity.

EDUCATION

Vishwakarma Institute of Technology, Pune

2023 – 2027

B.Tech in Computer Engineering | CGPA: 8.57 / 10 (as of 6th Semester)

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, SQL, JavaScript, HTML/CSS, Bash
- **Machine Learning & Computer Vision:** TensorFlow, Scikit-learn, OpenCV, CNNs, YOLO, NumPy, Pandas
- **Cybersecurity & Systems:** Linux, Anomaly Detection (Isolation Forest), Risk Scoring, Log Analysis & Correlation
- **Cloud & DevOps:** AWS (EC2, S3, IAM, VPC, Lambda, DynamoDB, EKS, ECS, ECR), Docker, Kubernetes, Git/GitHub
- **Core CS Fundamentals:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Junior AI Developer Intern | Dentsu Global Services

2 Months

- Developed CatalystM, an AI-powered Marketing Mix Modeling (MMM) tool to support automated marketing analytics and spend-performance evaluation.
- Built the application's full stack using React for the frontend, Python for the modeling/backend logic, and SQLite for data storage.

AI & Bioinformatics Intern | GeneSpectrum Life Sciences

1 Month

- Explored AlphaGenome and researched OpenCRAVAT for genomic variant annotation and interpretation as part of applied bioinformatics analysis.
- Developed an AI-powered RareVarDB chatbot, experimenting with the OpenRouter API and locally hosted Ollama open-source language models.
- Set up and explored the ClawBio pipeline for genomic analysis and clinical variant reporting.
- Prepared and presented a session on “Next Generation AI in Bioinformatics,” covering emerging AI technologies and their impact on life sciences.

Research Intern | IEEE EMBS Pune Chapter

1 Month

- Contributed to the project “Multistage Diabetic Retinopathy Detection Using Deep Learning” under faculty mentorship, applying deep learning to medical image analysis.
- Strengthened understanding of AI applications in healthcare while working on a real-world biomedical problem, with progress presented at the IEEE EMBS Internship Valedictory Program.

PROJECTS

Multistage Diabetic Retinopathy Detection using Deep Learning

Computer Vision, Machine Learning, Healthcare AI

- Developed a diabetic retinopathy detection model using ResNet50 in TensorFlow for automated retinal image classification and medical screening.
- Engineered an end-to-end deep learning pipeline with image preprocessing, augmentation, transfer learning, and class-imbalance handling across a dataset of 3,662 retinal fundus images.
- Achieved 93% validation accuracy, evaluated using precision, recall, F1-score, Cohen's Kappa, confusion matrix, and Grad-CAM visualization for model explainability.
- Built automated inference and prediction modules to enable immediate retina image analysis.

CPAM - Continuous Privilege Access Management Platform

Industry-Sponsored Project, in collaboration with Barclays | Cybersecurity, Linux, Machine Learning

- Designed a centralized, real-time privileged-access monitoring system to track and correlate authentication, Linux, and command-level activity across enterprise environments, with progress validated through periodic review meetings with Barclays stakeholders.
- Built a Python-Flask risk-analysis pipeline featuring session tracking, log normalization, and rule-based risk scoring for proactive threat monitoring.
- Integrated ML-based anomaly detection (Isolation Forest) and AI-driven insights (Ollama Mistral) to identify suspicious privilege-user behavior and generate actionable security alerts.
- Built an interactive monitoring dashboard (HTML, CSS, JavaScript, SQLite, Ubuntu Linux) for real-time visualization, alerting, and security analytics.

TransportBench - Real-Time Benchmarking Framework for Network Protocols

Computer Networks, Performance Profiling

- Built a full-stack framework to benchmark TCP, UDP, and QUIC protocol performance through real-time, socket-level experiments.
- Implemented custom socket programming and live dashboards to visualize latency and throughput metrics dynamically across multiple test scenarios.
- Surfaced protocol-level performance trade-offs to guide protocol selection for latency-sensitive and high-throughput applications.

ACHIEVEMENTS & LEADERSHIP

- **Smart India Internal Hackathon (SIH) – Top 50 of 900+ Teams:** Led team “Fast and Curious” to a Top 50 finish among 900+ submissions at VIT Pune; architected “BetterME,” a gamified mental-health monitoring platform featuring breathing exercises, focus tracking (tab-switch detection), and sleepiness analysis.
- **Hack4Health (BJ Medical College × COEP) – Runner-Up:** Led a cross-disciplinary team of medical and engineering students (“Code Cure”) to a runner-up finish; built “VentiSure,” a real-time VAP diagnosis and monitoring system integrating ventilator data, patient feedback, and PV-curve analysis for anomaly detection.
- **Chairperson, IEEE EMBS Student Chapter, VIT Pune:** Lead and manage the Engineering in Medicine and Biology Society chapter, organizing technical events and initiatives; collaborate with industry professionals from Genespectrum and Philips Healthcare and coordinate with the IEEE Pune Section.

CERTIFICATIONS

- **Professional & Cloud:** IBM Full Stack Developer Professional Certificate; AWS Cloud Technical Essentials (Amazon Web Services)
- **AI, Healthcare & Programming:** AI for Healthcare Systems (University of Colorado Boulder); Biology Meets Programming; Bioinformatics for Beginners (UC San Diego); Python & C Programming Certification (MCCE)